

Mixed Reality Object Viewing: What do People Remember about the Qualities of Real and Virtual Objects Viewed through Passthrough?

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Figure 1: A participant viewing the following objects through the Apple Vision Pro Headset, as part of the mixed reality source monitoring study: a virtual milk jug, a real mug, a virtual picture frame, and a virtual rolling pin.

ABSTRACT

The aim of this study was to evaluate the characteristics of memories of everyday objects viewed through the Apple Vision Pro mixed reality headset. 32 participants viewed 20 objects that were either real objects viewed in passthrough, or highly detailed 3D scans of real objects that appeared in the participants' environment. One week after viewing, participants were interviewed about their experience and asked to recall features of the objects that helped them determine whether they were real or virtual. A rapid qualitative analysis revealed 4 themes related to the nature of participant object memories: 1) Ease or challenge of distinguishing real and virtual, 2) the use of cues related to lighting, shadows, color, and visual texture 3) the nature of the objects and whether they were unusual or ordinary; and 4) the use of tactile exploration. Future work can utilize these features to enhance the memorability of virtual.

Index Terms: Mixed Reality, Memory, Passthrough, Qualitative

1 INTRODUCTION

There has been a remarkable public adoption of immersive technologies over the past four to six years, including the highly anticipated Apple Vision Pro. This mixed reality (MR) technology enables a novel "passthrough effect," allowing users to see their actual physical surroundings while engaged in digital content, greatly enhancing the realism of the experience. Greater realism opens the possibility that a user might confuse a virtual experience with a real one, or more specifically, might remember a virtual object seen in passthrough MR as a real object [5]. This question, whether someone can accurately recall the context in which something was experienced, has a long history of investigation in memory research through source monitoring paradigms [6, 7]. As immersive technology continues to become increasingly mainstream, there is a need to evaluate the current state of mixed realities' effect on user memory. Therefore, this study was designed as a replication-extension study that leverages

Hoffman et al.'s "Virtual Reality Monitoring" paradigm to assess how accurately people are able to remember the "source" of an encountered object: whether the object was a real physical object, or a virtual one overlaid on the real environment using passthrough MR technology [5]. After viewing a series of real and virtual objects, participants waited one week and then attempted to recall whether each object seen was real or virtual. Semi structured interviews were also conducted to probe participants' memories of the specific features and experiences recalled from their encounter with the objects. These interviews were analyzed using rapid techniques in order to uncover what qualities most contribute to the sense of "realness" of MR objects and how those features are recalled after a 1-week delay.

2 METHODS

Thirty-two participants (M age = 22.0 years, SD = 1.98; 19 females) wore the Apple Vision Pro Headset 3660x3200 resolution per eye, with 120Hz refresh rate; [1]) and viewed a series of objects. All participants had normal or corrected to normal vision. Individuals who were unable or unwilling to wear contact lenses (as opposed to glasses) to correct their vision were excluded from participation due to incompatibility with the Apple Vision Pro Headset. Participants were informed that they would need to distinguish real objects from virtual objects, and that they would participate in a follow up study 1 week after their initial participation, but they were not told that the study was about memory or source monitoring. All participants provided their informed written consent according to a protocol approved by Yale University's Human Subjects Committee.

2.1 Stimuli

Thirty real household objects were selected for the source monitoring task. An Arctec Space Spider scanner and Artec Studio 15 were used to create the high-fidelity 3D models and textures of the 30 real objects [2].

2.2 Procedure

During the study phase, participants were presented with 20 of the items. Each item appeared one at a time, at approximately arm's length, centered on a small table. Real objects were placed manually by the experimenter and could be viewed by the participant using the headset's passthrough feature. Virtual objects were 'anchored' to the horizontal plane of the table's surface such that they appeared to

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rest on the table and remained stationary even when the participant moved their head or leaned forward or backward. Participants were encouraged to reach out and touch the object during viewing to confirm whether it was real or virtual. Participants then received a \$25 gift card, and a follow up study session was scheduled. The follow up session was completed remotely over Zoom videoconferencing software. Participants were administered the Source Identification Test, in which they classified all 30 objects as “real,” “virtual,” or “new” and rated their confidence in each response. Finally, a brief structured interview was conducted in which participants were asked to describe their experience using MR and to provide an accounting of what cues were utilized to recount the source of the objects they saw during the in-person testing phase. Participants were paid an additional \$25 following the interview.

3 RESULTS

We used a rapid analysis technique based on Gale et al.’s adaptation of Hamilton’s rapid qualitative analysis approach [3,4]. This involved creating a summary of each interview shortly after completion, discussing notable interview outcomes after summarizing, and finally transferring summarized findings to a matrix. We created the matrix from keywords that recurred across interview summaries, such as color and lighting, and broader concepts related to interview probes, such as “distinguishing real objects from virtual objects.” As more interviews were conducted, we added sub-codes to add further structure to the matrix. There emerged 4 themes related to participants’ experience and memories of the virtual and real objects viewed: 1) Ease or challenge of distinguishing real and virtual, 2) the use of cues related to lighting, shadows, color, and visual texture 3) the nature of the objects and whether they were unusual or ordinary; and 4) the use of tactile exploration to check assumptions about realness of objects. Table 1 shows the counts of quotes associated with each theme.

Table 1: Quotes Related to Themes by Subtheme

Theme	Subtheme	Number of Quotes
Distinguishing real and virtual objects	Easy	11
	Challenging	13
Object characteristics	Lighting and shadows	7
	Color	13
	Texture	11
Nature of the objects	Unusual	6
	Ordinary	4
Tactile interaction	Determine real or virtual	9
	Touched out of curiosity	3

3.1 Theme 1

Thirteen participants (40%) noted that it was difficult to tell the difference between real and virtual objects, with many participants expressing surprise at the challenge:

“I was really surprised at how difficult it was to distinguish real and virtual objects by sight alone. It’s a combination of the virtual objects looking very real and the real objects having some grain, making them look faker.” – Participant 12

Participants who found the task easy (11; 34%) referred to specific strategies for identifying virtual objects, such as checking to see whether the object was in obvious contact with the table:

“I was sure that almost all the virtual ones were virtual because how “grounded” the objects seemed. Basically, I could tell they were not real because they were levitating a few centimeters off the table.” – Participant 16

3.2 Theme 2

Eleven participants (34%) remarked about the visual texture of the objects, many mentioning that the virtual objects were more detailed than expected:

“I thought [the tennis ball] was real, but it wasn’t. I think it’s like the texture of it, and also I could see the like shades, like the gradient. I don’t know if that’s the word. The fuzz.” – Participant 32

Eleven participants (34%) commented that all the objects (both real and virtual) seemed brighter and more saturated in color than expected. Two additional participants said that the virtual colors were “off” compared to the real objects’ coloring, and 7 participants specifically discussed the shadows cast by the objects, or the way the virtual objects reflected light and how the reflection did not correspond to the lighting conditions of the room:

“The [TV remote] controller looked super real at first glance, but then the white buttons caught my eye. They were unusually bright and white—almost glowing—which tipped me off that it might be virtual.” – Participant 15

“The ones that did look virtual, I remember like they’re just like a little like unnaturally colored or like shaded or like the lighting of it didn’t look like it would be in that room right now with the overhead lighting.” – Participant 27

3.3 Theme 3

Ten participants (31%) commented on the expected or unexpected nature of the items and how that contributed to their ability to remember the item and distinguish real objects from virtual objects. Participants expressed a tendency to guess more “ordinary” objects were real objects, and more unusual or unique objects were virtual:

“I used my familiarity with the objects; a lot of my pre-suppositions were about what real world objects looks are and preconceived what was expected. Like is this like what a bowl would look like versus a .jpg image of a bowl?” – Participant 18

“I often found that I really noticed a specific or weird object, like the pufferfish [toy], but I got confused on whether I remember it because it was obviously fake or because I touched it and felt that is was real.” – Participant 13

3.4 Theme 4

Although all participants were told they could touch each object as it appeared in front of them, only 9 participants (28%) explicitly stated that they attempted to touch objects to confirm whether or not they were real:

“Yeah, I’m pretty sure I didn’t reach out to every object cause some of them, I think I just like knew that it was virtual, but then the ones that I couldn’t tell, I would reach for it. What I remember, I think there was like a vase or something...I think I tried to reach out for it, but I couldn’t feel anything. So it was virtual.” – Participant 21

“I remember when it was first placed in front of me, I didn’t expect it to be real at first. I was like, ‘oh, wow, that’s a pizza box. Wow.’ So I like wanted to open it, and I was kind of surprised that it, you know, lifted up.”
– Participant 20

4 DISCUSSION

The themes that emerged from our rapid content analysis reflect the characteristics of objects in the study that remained salient to participants after seven days had elapsed. The delay between stimuli exposure and structured interview, necessary for the source monitoring paradigm, likely resulted in diminished memory traces for all participants and yet many specific details were discussed during interviews. Broadly speaking, participants who commented on the ease of distinguishing real objects from virtual conveyed a specific strategy or quality of the objects that they used to make their determination. Participants who thought the task was hard did not offer details they looked for, although many commented on standout object characteristics, such as lighting and shadows, color, or visual texture. Many of the most detailed memories recounted during the interviews were about the unexpected or familiar nature of the objects used in the study: participants stated that some of the small toys used were nostalgic or made them smile, and that this was unexpected. On the other hand, many participants commented that the objects were extremely ordinary and that it made it hard to recall them specifically. Touching objects was explicitly allowed by the experimenter, who often reminded participants to give it a try. Despite these instructions, not all participants attempted to touch items and very few mentioned their tactile interactions with objects during the interview.

5 CONCLUSION

The findings from this study aim to enhance our understanding of how immersive technologies affect memory processes, potentially informing the design and application of safer and more effective MR environments by identifying the features of digital objects most likely to be misremembered and cause source confusion between real and virtual objects.

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